David Song

San Francisco Bay Area, CA | (669) 236 - 1331 | davidhsongg@gmail.com | Portfolio | LinkedIn | GitHub

Education

San Jose State University — San Jose, CA
B.S. in Computer Science, Minor in Digital Media Art — Aug 2021 – Dec 2025

Experience

Software Engineer Intern — May 2024 – Jul 2024
Life Stages (lifestages.us) — San Francisco, CA

- Halved the React Daily Check-in flow from 6 steps to 3 and added validation to prevent invalid submissions
- Connected check-in and profile screens to REST APIs and defined request and response data with TypeScript
- Replaced repeated interface elements with reusable React components, simplifying updates across screens
- Fixed an iOS Safari flexbox issue that prevented the check-in form from displaying correctly
- Added Jest unit and integration tests and GitHub Actions checks for coverage

Selected Projects

Wi-Fi CSI-Based Activity Classification — Oct 2026 – Present
Python, PyTorch, NumPy, pandas

- Trained a PyTorch model to recognize 3 activities from 461 Wi-Fi signal samples
- Measured accuracy and F1 score on test data across 2 antenna designs and clear or blocked Wi-Fi paths
- Added scripts to train the model, save checkpoints, and classify a new Wi-Fi sample

Google Business Profile Review Sync and Reply Drafting — Jan 2026 – Jun 2026
Next.js, TypeScript, Supabase/PostgreSQL, Stripe, Claude API, AWS Lambda, EventBridge, CloudWatch

- Built an installable review platform tested by 2 businesses; synced Google Business Profile reviews and replies
- Drafted review replies in each business's voice with Claude and labeled reviews by sentiment and urgency
- Added Stripe subscriptions and Resend email digests; scheduled review syncs with AWS Lambda and EventBridge

Mobile Car Detailing Booking and Availability Management — Sep 2025 – Jan 2026
Next.js, TypeScript, PostgreSQL, Supabase, AWS Lambda, EventBridge

- Built and operated a mobile booking platform supporting 26 completed bookings and $3,900 in revenue over 2 months
- Set separate customer, operator, and admin access rules; automated expired booking holds and availability updates
- Added rate limits and account lockout; used CSP, HSTS, and frame protection on booking and admin pages

E-Commerce Product Search and Order Management — Jan 2025 – Apr 2025
React, Node.js, Express.js, Elasticsearch, PostgreSQL, Google OAuth 2.0

- Built a React storefront with product filters and live cart totals
- Added live typo-tolerant Elasticsearch search with custom analyzers and fuzzy matching
- Built 22 REST endpoints and admin tools for products, orders, and payments with JWT and role-based access

IoT Environmental Monitoring Dashboard — Oct 2023 – Nov 2023
Next.js, Python, Raspberry Pi, Firebase, Google Sheets API

- Led a 3-person frontend team to build a dashboard for live and historical temperature and humidity monitoring
- Built a Raspberry Pi Python service that read I2C sensors every 60 seconds and stored readings in Firebase
- Added Google sign-in, Drive file selection, and Google Sheets exports for sensor data

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, Express.js, FastAPI, PyTorch, Jest, React Testing Library
Databases & Search: PostgreSQL, MySQL, SQLite, Redis, Elasticsearch, Supabase, Firebase
Cloud & DevOps: AWS Lambda, EventBridge, CloudWatch, Docker, Git, GitHub Actions, Vercel
APIs, AI & Security: REST APIs, Anthropic Claude API, OAuth 2.0, JWT, Row Level Security, Stripe, Resend